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[Q1]: Protein of Interest

Protein: Kinesin Family Member 11 (KIF11)

Species: Homo sapiens

Accession Number: NM_004523.4 (RefSeq protein)

Function: KIF11 is a kinesin motor protein that helps separate chromosomes during mitosis by organizing and maintaining the bipolar spindle. It uses ATP-powered movement along microtubules to support proper cell division.

[Q2]: BLAST Search

A BLASTN search was performed using the nucleotide sequence of the human KIF11 gene (accession: NM_004523.4) against the Transcriptome Shotgun Assembly (TSA) database limited to bacteria (taxid:2).

Organism: Bacteria

“Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i5”
(Accession: GFOJ010595215.1)

- **Max Score: 344**
- **E-value: 2e-90**
- **Percent Identity: 68.06%**
- **Query Coverage: 21%**

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Your search is limited to records include: Bacteria (taxid:2)

Job Title

NM_004523:Homo sapiens kinesin family member...

RID

0TBN955D014

Search expires on 05-21 02:40 am

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Database

tsa (25 databases)

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Query ID

NM_004523.4

Description

Homo sapiens kinesin family member 11 (KIF11), mRNA

Molecule type

nucleic acid

Query Length

5016

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to

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	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☐	Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i5	344	344	21%	2e-90	68.06%	3425	GFOJ010595215.1
☐	Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i4	344	344	21%	2e-90	68.06%	2358	GFOJ010595214.1
☐	Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i6	344	344	21%	2e-90	68.06%	3303	GFOJ010559821.1
☐	Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i3	344	344	21%	2e-90	68.06%	2300	GFOJ010559812.1
☐	TSA: Rhodococcus rhodochrous TRINITY_DN1599_c4_g1_i1 transcribed RNA sequence	262	340	6%	6e-66	80.78%	317	GGWF01000084.1

Visual pairwise alignment:

TSA: Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i5, transcribed RNA sequence

Sequence ID: GFOJ010595215.1 Length: 3425 Number of Matches: 1

Range 1: 2106 to 3142 GenBank Graphics

Next Match Previous Match

Score	Expect	Identities	Gaps	Strand
344 bits(381)	2e-90	714/1049(68%)	23/1049(2%)	Plus/Minus
Query 495	AAGTGTGTTTGTCTCAATTCTGGATGAAGTTATTATGGGCTATAATTGCACTATCTTTGC	554		
Sbjct 3142	AAGTATTGTGTACCAATAATTAGTGAAGTCTGAGTGGTTACAACGTACAATTTTGC	3083		
Query 555	GTATGGCCAACTGGCACTGGAAAACTTTTACAATGGAAGGTGAAAGGTACACCTAATGA	614		
Sbjct 3082	CTATGGCCAAACAGGGACCGGAAAAACCTTCACTATGGAAGGGGAGCGATCAG---ATGA	3026		
Query 615	AGAGTAT---ACCTGGGAAGAGGATCCCTTGGCTGGTATAATTCACGTACCCTTCATCA	671		
Sbjct 3025	CCCCTGTGTACCTGGGAAGATGACCCCTTACTGGTATCATCCCAAGAGCGCTGGCCAA	2966		
Query 672	AATTTTTGAGAACTTACTGATAATGGTACTGAATTTTCAGTCAAAGTGTCTCTGTTGGA	731		
Sbjct 2965	CATATTGTATGTTCTCACCACAGGAGGTGGAATCTCCGTCGAGTGTCTTTCTGGA	2906		
Query 732	GATCTAATGAAGAGCTTTTGTATCTTCTTAATCCATCATCTGATGTTTCTGAGAGACT	791		
Sbjct 2905	ACTCTACAATGAAGAGCTGTTTGTATCTGCTTGGTTCATCAGAAGATCCA-CTGA-AGTTG	2848		
Query 792	ACAGATGTTTGTATGATCCCCGTAACAAGAGAGG-----AGTGATAATTAAAGGTTTAGAAG	847		
Sbjct 2847	AGA-ATATACGAAGA---CAGTG-CAAGGAAGGGCTCAGTAATAATCCATGGCCTAGAAG	2793		
Query 848	AAATTACAGTACACAACAAGGATGAAGTCTATCAAATTTTAGAAAAGGGGGCAGCAAAAA	907		

The selected hit has an E-value of 2e-90, a score of 344 bits, and 68% identity across the aligned region.

[Q3]: Novel Protein

Selected novel protein hit: Candidatus Thiodiazotropha endolucinida
Codakia_V2_299248_c0_g4_i5

Species: Candidatus Thiodiazotropha endolucinida

FASTA Sequence (translated ORF using EMBOSS Transeq):

```
>GFOJ010595215.1_5 TSA: Candidatus Thiodiazotropha endolucinida  
Codakia_V2_299248_c0_g4_i5  
EIFLQKI*EGLKQNLKSMPSQQKKEKSDQHIQVAVRLRPINAVEKKQCSYSVVDGSADRR  
EVTVKERLGINPTTKTFNFDHVFPEVKQVQIYKSIVSPIISEVLSGYNCTIFAYGQTGT  
GKTFTMEGERSDDPCVTWEDDPLTGII PRALANIFDVLTNQEVEFSVRVSFLELYNEELF  
DLLGSSDPLKLRIYEDSARKGSVIIHGLEEMVVRSKDEVYDILERGAARRQTAATLLNA  
HSSRSHSVFTVTIHIKENTMEGEELLKTGKLHLVDLAGSENIGRSGAVDKRAREAGNINQ  
SLLTLGRVITALVEHAPHIPYRESKLTRLLQDSLGGRTKTSIIATVSPASINLDETLSTL  
DYAHRAKNITNRPEINQKLTKRALIKETEEIEKLRRDLQASREKNGFYIAEDNFNSMQM  
KLTVQQENIEEMTEKIEVLSDDELTKTQELFSDTKKDLEQTTEKLTETTENLRNTKVTLKE  
TKITLRDTEQDRNEQRFLVDEHVKTETDLYSQATELLGTVGESTNDVVRGLHEKVDRLKAV  
ESHNEECQGSFQRKYHQDIQSMMDELKGFLENEQETSHNAHTEHIAQKVAKNKHVSELTS  
EVIELQKGLQERVESMNSQQKTCACNSMSWMKDLKNSYLTKEEEIQSIGKFEEEMISKS  
TAVVSKLQNMTESLDQLSSLIEQQKQEEEEQRLQNWADRLTGDLDKLNSLVTQYSSRQLET  
MTGLKTHVVDQTQTEYNATKQEIDVKMNEIFQIADKLKELNTLLAKNSSTVTQHSNHTLSE  
IEKSSSSAEDMSSNLNSEINHIKERSNSCLQESQEKLESNCQCVHKDIDKIKSVNTTTVD  
IQTEIQTYNTVGKQQRIKFLDSYQDTLEDRCTEHETTINTDLNVFYKNTDALQSENQSAC  
DKLISSLDSTASFDEAVATENQLLISRISGVRNFSQDQSARLTDRSDVQRFVGEELRK  
DIPTGQTPVRKDFQYPRDLKKTGRHSLLEKFREDNMVNLSLDEDSIVEESDEIKETEMQ  
NDDVADGKG*CC*KLLNIRCIFSISPQ*G*GYRISGK*KTRSTPGI*KKSEKKFQKCY  
TKETDQTATENSYAKWTVKLNFRVNLLQIKQ*RKRLWWIFGDNWIIIFSSSP*NYIF*VL  
IX
```

[Q4]: Proving Novelty

A BLASTP search of the translated protein sequence against the nr database showed that the top match (kinesin-like protein KIF11-B [Mya arenaria]) had 54.66% amino acid identity to a protein from a different species, Mya arenaria. To conclude: this protein is likely novel because there is no 100% identical match from the same species.

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GFOJ010595215.1_5 TSA: Candidatus Thiodiazotropha...

RID

OTCS8XVE016

Search expires on 05-21 02:59 am

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Database

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Query ID

lcl|Query_1034714

Description

GFOJ010595215.1_5 TSA: Candidatus Thiodiazotropha en...

Molecule type

amino acid

Query Length

1142

Other reports

Distance tree of results

Multiple alignment

MSA viewer

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	Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
	Click the icon to see the cluster contents									
☒	1 member(s), 1 organism(s)	softshell	kinesin-like protein KIF11-B [Mya arenaria]	1038	1038	87%	0.0	54.66%	1094	XP_052770351.1
☒	1 member(s), 1 organism(s)	bivalves	unnamed protein product [Chamelelea gallina]	1031	1031	87%	0.0	53.14%	1077	CAN1469806.1
☒	1 member(s), 1 organism(s)	northern quahog	kinesin-like protein KIF11-B [Mercenaria mercenaria]	1026	1026	88%	0.0	52.72%	1077	XP_053406875.1
☒	1 member(s), 1 organism(s)	Manila clam	kinesin-like protein KIF11 [Ruditapes philippinarum]	994	994	88%	0.0	51.89%	1090	XP_060567348.1
☒	6 member(s), 1 organism(s)	zebra mussel	hypothetical protein DPMN_163994, partial [Dreissena polymor...	984	984	87%	0.0	51.98%	1129	KAH3785899.1
☒	2 member(s), 2 organism(s)	bivalves	hvothetical protein ACJMK2_005000 [Sinanodonta woodiana]	983	983	86%	0.0	52.65%	1086	KAL3863235.1

Visual pairwise alignment:

kinesin-like protein KIF11-B [Mya arenaria]

Sequence ID: [XP_052770351.1](#) Length: 1094 Number of Matches: 1

Range 1: 1 to 985 [GenPept](#) [Graphics](#)

Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps
1038 bits(2684)	0.0	Compositional matrix adjust.	551/1008(55%)	714/1008(70%)	39/1008(3%)
Query 18	MPSQQKKEKSDQHIQVAVRLRPINAVEKKQCSYSVVDGSDRREVTVKERLGINPTTKTF	77			
Sbjct 1	MP + ++ +QHIQVAVRLRPIN+ E+KQ SYSVVDG+A+++EVTVKERLG+NPTTKTF	60			
Query 78	NFDHVFPEVKQVQIYKSIVSPIISEVLSGYNCTIFAYGQTGTGKFTMEGERSDDPCVT	137			
Sbjct 61	NFDHVFPPQDVKQQVYKSIVSPIIDEVLGDGYNCTIFAYGQTGTGKFTMEGERSDDPNMS	120			
Query 138	WEDDPLTGIIPRALANIFDVLTNQEVFVSRVVSFLELYNEELFDLLGSSDPLKRIYED	197			
Sbjct 121	WEDDPLTG+IPRALANIFD L Q VEFVSRVVSFLELYNEELFDLLGSS DPL+L+IYED	180			
Query 198	SARKGSVIIHGLEEMVVRSKDEVYDILERGAARRQTAATLLNAHSSRSHSVFTVTIHIKE	257			
Sbjct 181	S RKGSVII GLEE+VV+SKDEVY+ILERGA+RRQTAATLLNA SSRSHSVFTVTIH KE	240			
Query 258	NTMEGEELLKTGKLHLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVEHAP	317			
Sbjct 241	NTMEGEELLKTGKLHLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVEHAP	300			
Query 318	HIPYRESKLTRLLQDSLGGRTKTSIIATVSPASINLDETSLTDYAHRAKNITNRPEINQ	377			
Sbjct 301	HVPYRESKLTRILQDSLGGRTKTSIIATISPASVNLEETSLTDYAHRAKNITNRPEVNQ	360			

The selected hit has an E-value of 0, a score of 1038 bits, and 55% identity across the aligned region.

[Q5]: Multiple Sequence Alignment

A multiple sequence alignment was generated using Clustal Omega. The alignment includes the original human KIF11 protein query, the novel *Candidatus Thiodiazotropha endolucinida* protein, the top BLASTP hit (*Mya arenaria* KIF11-B), and five additional KIF11 family homologs from different species.

CLUSTAL O(1.2.4) multiple sequence alignment

Human_KIF11	-----MASQPNSSAKKKEEKGNIQVV	22
Sinanodonta_woodiana	-----MP-----QPKVNNNQNIQVA	15
Novel_Thiodiazotropha	-----EGLKQNLKSMPS-----QOKKEKSDQHIQVA	26
Ruditapes_philippinarum	-----MPP-----LPKA--KQENQHIQVA	17
Chamelea_gallina	-----MPP-----LPKVKDKQENQHIQVA	19
Mercenaria_mercenaria	-----MPP-----LPKVKDKQENQHIQVA	19
Mya_arenaria	-----MPP-----PKGCKQ---ENQHIQVA	17
Dreissena_polymorpha	VAPDVLVVFKESTRRLHASGTLINSTRRLVRLVVRVAPDVLIVFEGSTRRLHASGTLVNNTR	60
Human_KIF11	VRCRPFNLAERKASAHSIVECDPVRKEVSVRTGGLADKSSRKYTFDMVFGASTKQIDVY	82
Sinanodonta_woodiana	VRLRPINSIERKQGSYSVVDSENGEKEVVIKE-KLGVNPSTRTFNFDHVFPPYSKQIDLY	74
Novel_Thiodiazotropha	VRLRPINAVEKKQCSYSVVDGSADRREVTVKE-RLGINPTTKTFNFDHVFPEVKQVQIY	85
Ruditapes_philippinarum	VRLRPINSSERKQGSYSVADANSLRKEVTVKE-RLGVNPSTKTFNFDHVFQDSKQQDVY	76
Chamelea_gallina	VRLRPINSSERKQGSYSVADASGEKKEVTVKE-RLGVNPSTKTFNFDHVFQDSQQQVY	78
Mercenaria_mercenaria	VRLRPINSSERKQGSYSVADASSEKREVTVKE-RLGINPTTKTFNFDHVFQDSKQQQVY	78
Mya_arenaria	VRLRPINSTERKQSSYSVVDGNAEKEVTVKE-RLGVNPSTKTFNFDHVFQDVKQQQVY	76
Dreissena_polymorpha	RLFRPINSARERKQGSYSVVEANSEKKEVNVKE-RLGINPTTKTFNFDHVFQDVKQTDVY	119
Human_KIF11	RSVVCPIILDEVIMGYNCTIFAYGQTGTGKFTMEGERSPNEEYTWEDPLAGIIPRTLHQ	142
Sinanodonta_woodiana	KTIVTPIIIEVLIMGYNCTIFAYGQTGTGKFTMEGERTDDPTLTWDDPLSGIIPRAMAN	134
Novel_Thiodiazotropha	KSIVSPIIEVLSGYNCTIFAYGQTGTGKFTMEGERSDDPCVTWEDDPLTGIIPRALAN	145
Ruditapes_philippinarum	KTVVSPIIIEVLIMGYNCTIFAYGQTGTGKFTMEGERSDDPLTWEQDPLTGIIPRSLAN	136
Chamelea_gallina	KTVVSPIIAEVLIMGYNCTIFAYGQTGTGKFTMEGERSSENTNLWEQDPLTGIIPRALAN	138
Mercenaria_mercenaria	KTVVSPIIAEVLIMGYNCTIFAYGQTGTGKFTMEGERSENSELTWEQDPLTGIIPRALAN	138

Mya_arenaria	KSIVSPIIDEVLGDYNTIFAYGQTGTGKTFTMEGERSDDPNMSWEDDPLTGVI PRALAN	136
Dreissena_polymorpha	KSIVSPIIKEVLGDYNTIFAYGQTGTGKTFTMEGERSDPNLTWENDPLTGI I PRSLAN	179
Human_KIF11	IFEKLTNGTEFSVKVSLLEIYNEELFDLLNPFSSDVSERLQMFDDPRNKRGVIIKGLEEI	202
Sinanodonta_woodiana	IFDTLQSQEVEFSIRVSFIELYNEELFDLLGSLED-TLRLKIYEDNARKGSVIIQGLEEI	193
Novel_Thiodiazotropha	IFDVLTNQEVEFSVRVSFLELYNEELFDLLGSSED-PLKLRIYEDSARKGSVIIHGLEEM	204
Ruditapes_philippinarum	IFDSLQSQECEFSVRVSFLELYNEELFDLLGSSID-PLRLKIYEDSSRKGSVIIQGLEEI	195
Chamelea_gallina	IFDNLQSQECEFSVRVSFLELYNEELFDLLGSSVD-PLRLKIYEDTSRKGSVIIQGLEEI	197
Mercenaria_mercenaria	IFDTLQSQECEFSVRVSFLELYNEELFDLLGSSVD-PLRLKIYEDTSRKGSVIIQGLEEI	197
Mya_arenaria	IFDQLQAQAVEFSVRVSFLELYNEELFDLLGSSID-PLRLKIYEDSTRKGSVIIQGLEEI	195
Dreissena_polymorpha	IFDQLQSQEVEFSVRVSFLELYNEELFDLLGSSVD-PLRLKIYEDSTRKGSVIIQGLEEI	238
Human_KIF11	TVHNKDEVYQILEKGAARKRTTAATLMNAYSSRSHSVFSVTIHMKETIDGEELVKIGKLN	262
Sinanodonta_woodiana	VVRSKEEVYEILERGAARRQTAATLMNAHSSRSHSVFSVTIHIKENTMDGEELLKTGKLY	253
Novel_Thiodiazotropha	VVRSKDEVYDILERGAARRQTAATLLNAHSSRSHSVFTVTIHIKENTMEGEELLKTGKLH	264
Ruditapes_philippinarum	VVKSDEVYEILERGASKRQTAATLMNAFSSRSHSVFTVTIHTKENTIDGEELLKTGKLH	255
Chamelea_gallina	VVKSDEVYEILERGASRRQTAATLMNAFSSRSHSVFTVTIHTKENTIDGEELLKTGKLH	257
Mercenaria_mercenaria	VVKSDEVYEILERGASRRQTAATLMNAFSSRSHSVFTVTIHTKENTIDGEELLKTGKLH	257
Mya_arenaria	VVKSDEVYEILERGASRRQTAATLLNAFSSRSHSVFTVTIHTKENTMEGEELLKTGKLH	255
Dreissena_polymorpha	IVKSDEVYEILERGASRRQTAATLLNAFSSRSHSVFTVTIHTKENTMEGEELLKTGKLH	298
Human_KIF11	LVDLAGSENIGRSGAVDKRAREAGNINQSLTLGRVITALVERTPHVPYRESKLTRILQD	322
Sinanodonta_woodiana	LVDLAGSENIGRSGAVDKRAREAGNINQSLTLGRVITALVEHAPHVPYRESKLTRLLQD	313
Novel_Thiodiazotropha	LVDLAGSENIGRSGAVDKRAREAGNINQSLTLGRVITALVEHAPHIPYRESKLTRLLQD	324
Ruditapes_philippinarum	LVDLAGSENIGRSGAVDKRAREAGNINQSLTLGRVITALVEHAPHIPYRESKLTRILQD	315
Chamelea_gallina	LVDLAGSENIGRSGAVDKRAREAGNINQSLTLGRVITALVEHAPHIPYRESKLTRILQD	317
Mercenaria_mercenaria	LVDLAGSENIGRSGAVDKRAREAGNINQSLTLGRVITALVEHAPHIPYRESKLTRILQD	317
Mya_arenaria	LVDLAGSENIGRSGAVDKRAREAGNINQSLTLGRVITALVEHAPHVPYRESKLTRILQD	315
Dreissena_polymorpha	LVDLAGSENIGRSGAVDKRAREAGNINQSLTLGRVITALVEHAPHIPYRESKLTRILQD	358
Human_KIF11	SLGGRTRTSIIATISPASLNLEETLSTLEYAHRANKNILNKPEVNQKLTKKALIKEYTEEI	382
Sinanodonta_woodiana	SLGGRTKTSIIATISPASSCLEETLSTLDYAFRAKNITNRPEINQKLTKKALIREYTEEI	373
Novel_Thiodiazotropha	SLGGRTKTSIIATVSPASINLDETSTLDYAHRAKNITNRPEINQKLTKRALIKEYTEEI	384

Ruditapes_philippinarum	SLGGRTKTSIIATISPASINLEETLSTLDYAHRAKDITNRPEVNQKLTKKALIKKEYNDEI	375
Chamelea_gallina	SLGGRTKTSIIATISPASINLDETSTLDYAHRAKNITNRPEVNQKLTKKALIKKEYNDEI	377
Mercenaria_mercenaria	SLGGRTKTSIIATISPASINLEETLSTLDYAHRAKNITNRPEVNQKLTKKALIKKEYNDEI	377
Mya_arenaria	SLGGRTKTSIIATISPASVNLEETLSTLDYAHRAKNITNRPEVNQKLTKKALIREYNEEI	375
Dreissena_polymorpha	SLGGRTKTSIIATISPASVNLEETLSTLDYAYRAKNITNRPEVNQKLTKRTLIKKEYTEEI	418
Human_KIF11	ERLKRDLAAAREKNGVYISEENFRVMSGKLTQEEQIVELIEKIGAVEEELNRVTELFMD	442
Sinanodonta_woodiana	EKLKRDHLAAREKNGIYLAEENYNAMHTKLTQQESINDMTEKITAMSEELNKLEIFKE	433
Novel_Thiodiazotropha	EKLRRDLQASREKNGFYIAEDNFNSMQMKLTQQENIEEMTEKIEVLSDELTKTQELFSD	444
Ruditapes_philippinarum	DKLRRDLQASREKNGIYIAEENYNMQMKITVQDDQIAELSDRIEAITTELSKVQDLFSE	435
Chamelea_gallina	DKLRRDLQAAREKNGFFIAEENYNMQKITASDDLIAELNDRIEALTEELTKVQDLFSE	437
Mercenaria_mercenaria	DKLRRDLQAAREKNGFFIAEENYNMQKITAQDDLIVELNDRIEAITDELAKVQDLFSE	437
Mya_arenaria	DKLRRDLQAAREKNGFFIAEENYNAMHVKLTAQDDQITELSTRIEAMTDELNVQELFAD	435
Dreissena_polymorpha	DKLRRDLQAAREKNGFFIAEENYNMQIKLTTQDDEIAELTRIEVMTEELHKITELFSD	478
Human_KIF11	NKNELDQCKSDLQNKQTQELTTQKHLQETKLQLV-----KEEYITSALESTEEKLHDA	495
Sinanodonta_woodiana	TEKNLEVTTQTLVTTTENLIKTSSELSHVTKQTLRETQKDRDEHKFLVDEHVKNETSLYGQ	493
Novel_Thiodiazotropha	TKKDLEQTTEKLTETTENLRNTKVTLKETKITLRDTEQDRNEQKFLVDEHVKTETDLYSQ	504
Ruditapes_philippinarum	TQEELQVTTDKLTLTKEKLVKTKTNLRETQVTLRQTEQDRNEQKFLVSEHLKNENELYEN	495
Chamelea_gallina	TQQELQQKTEKLAVTTDKLVKTKTNLRETQVTLRQTEQDRNEQKFLVSEHQKNEAQLYEN	497
Mercenaria_mercenaria	TQQELQVKTERLAVTTDKLVKTKTNLRETQVTLRHTEQDRNEQKFLVSEHQKNETQLYEN	497
Mya_arenaria	TRVELEQTAEKLTVTRDKLGKTRSTLHETKVTCLKTTELDKDRSHKYLQELTESQDTLFT	495
Dreissena_polymorpha	TKDKLEETSEKLTHTQDRLVTRTTTLKDTKVTCLKVTELELDGHRHLAKELTQSEGDLFDT	538
Human_KIF11	ASKLLNTVEETTKDVSGLSKLDKKAQVDQHNAEAQDIFGKNLNSLFNNMEELIKDGSSK	555
Sinanodonta_woodiana	ASVLLDTVQILHKDNEGLFAKIDRKKKVESHNEQSQQSFRRFRSEMDTMDNQLNQYLKT	553
Novel_Thiodiazotropha	ATELLGTVGESTNDVRGLHEKVDRKAVESHNEECQGSFQRKYHQDIQSMMDELKGLFNE	564
Ruditapes_philippinarum	ALSLNTAEVSTSDVSGLHEKLDKMSVEQQNKETVKLFEEFNENVSHIEEDAQSYKTQ	555
Chamelea_gallina	ATSLHTVDESTSDVTGLHEKVDRKMSVEQHNKTIVESYEECFNTGIGQMEQDGENYFKQ	557
Mercenaria_mercenaria	ATSLHTVDESTSDVVLGHEKLDKMTVEQHNKRTVESYEECFNTGLGQMEQDGESYLYKQ	557
Mya_arenaria	ANSLNTVDEATSDVSGLHDKVERLKTVEAHNLQTLDSYEETYNTSITTLTETCIQEYAEK	555
Dreissena_polymorpha	ASSLLQTVDRATSDVSGLHDKVARLKSVEAHNKVTLTQYEETYSSVVDMETYLRYAE	598

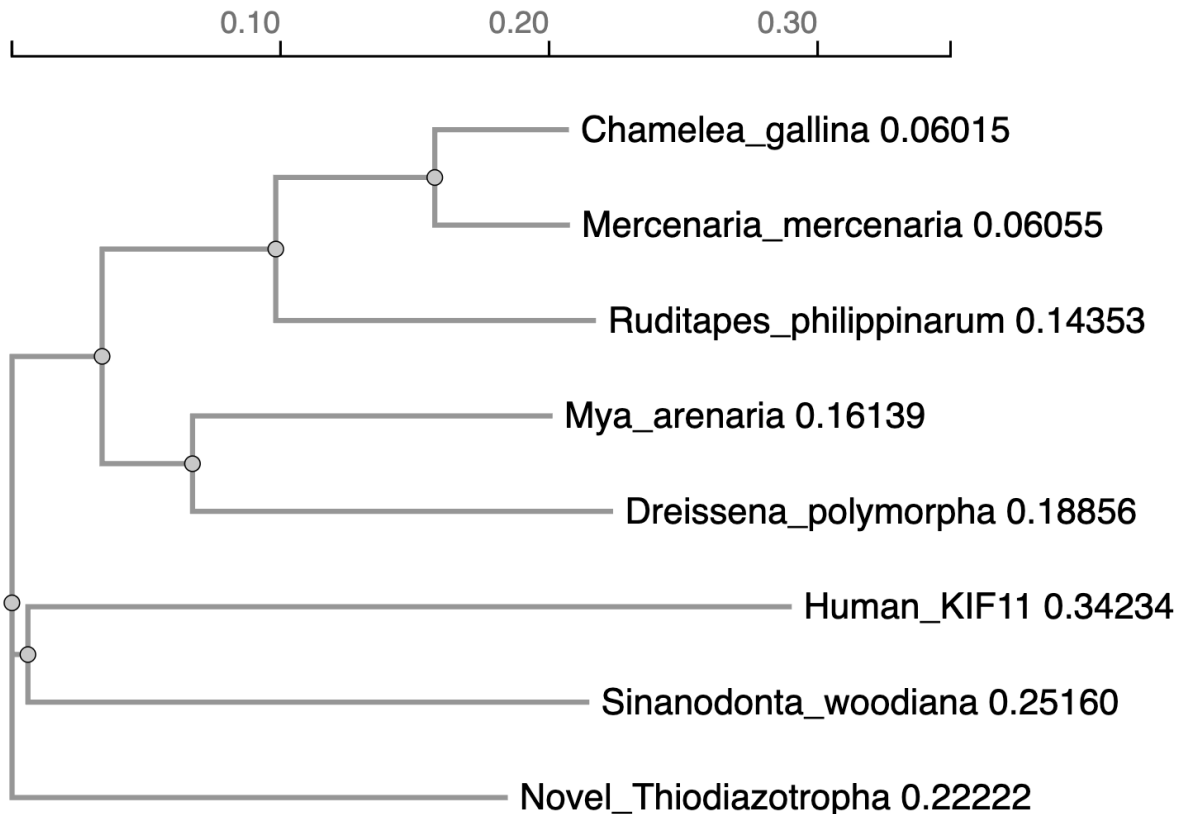
Human_KIF11	QKAMLEVHKTFLFGNLLSSSVSALDTITTTVALGSLTSIPENVSTHVSQIFNMILKEQSLAA	615
Sinanodonta_woodiana	QKAYHTKNAHSLMETKQRSEQAVKSLIDSIGGLQTHLTGHLR-----SLHDQQEECH	605
Novel_Thiodiazotropha	QETSHNAHTEHIAQKVAKNKHYVSELTSEVIELQKGLQERVE-----SMNSQQKTCA	616
Ruditapes_philippinarum	QODSHQTCLANIAKKVERSKEFTDNATTFLEFQTLVLDQLN-----VLSQHHKQQA	607
Chamelea_gallina	QQDIHQCTCTHISSKVKNKEFTDATTGHMRDFQTCINQQLD-----ALSQHHKQWT	609
Mercenaria_mercenaria	QODSHQTYTTHITCKVQRNKEFTDTTTTHMRDFQSCCLKQLD-----ELSQHHKQWT	609
Mya_arenaria	QAEVYAENSKHIAQKVERNKTYVSQLTQMRDLQSCLEQNT-----VISQHAKDTC	607
Dreissena_polymorpha	QARTYAQHTDHIASKITRNKDHVARLTDQMRELQSCLRDHVS-----AVTQSAVDGC	650
Human_KIF11	ESKTVLQELI---NVLKTDLLSSLEMLSPT-VVSILKINSQLKHIFKTSLTVADKIED	670
Sinanodonta_woodiana	DALVSWTNQVLDAANTCKVEELERLDIFHGNTIQTMTSVKSSINTLLETHNSVMQIVEK	665
Novel_Thiodiazotropha	CNSMSWMKDLKNSYLTKEEEIQSIGKFEEEM-ISKSTAVVSKLQNMTESLDQLSSLIEQ	675
Ruditapes_philippinarum	EENTSWVKVNSAASSLQDETLKKIEETKSIV-ETGISSLLQQMANISSSMPMKQLLDQ	666
Chamelea_gallina	QQQTSWVDGINTAAANMDRTVKKIEEIRNIV-NTGFTGLLAEMGNIGNSLDEVKQLVET	668
Mercenaria_mercenaria	QQQTSWVNGISSAATSFKDKTVNKIEEIRNLV-ESGFTSLLEEMGSISTSLDEVNQLLET	668
Mya_arenaria	EQNSCWANGISTGAADFKAESSRYEAIQGIL-GDKLTRAMTQLQDIGESHTQILELLET	666
Dreissena_polymorpha	EQNTAWVESVSIQAQQQLQEGENSRHKDIERVL-ADKLQGAMSEMAAIGASLGQIMQILES	709
Human_KIF11	QKKELDGFSLILCNNLHELQENTICSLVESQKQCGNLTEDLKTIKQTHSQELCKLMNLWT	730
Sinanodonta_woodiana	QAEQVATQEQAWEQQHQ-----LYDRLSQLVF	693
Novel_Thiodiazotropha	QKQEEEQRLQNWADRLTG-----DLDKLNSLVT	703
Ruditapes_philippinarum	QRAEQNTILGTVSDECS-----GLKDLGDTCE	694
Chamelea_gallina	QGKEQVEFLTTRSAQCST-----SLQELGDMYQ	696
Mercenaria_mercenaria	QGKEQVEILKTQSSQCSS-----SLQELGDMYE	696
Mya_arenaria	QRQEGETSVQAWSERLTK-----ELGDLVGLVE	694
Dreissena_polymorpha	QREAEVRQHAWAERLTQ-----SLCHLVSSVQ	737
Human_KIF11	ERFCALEEKCENIQKPLSSVQENIQQKSK-----DIVNKMTFHS----	769
Sinanodonta_woodiana	QYRDEQEKKITQLQQQLQEAQQTNSQLNKEAEIKFAELEQIFTKRNSLFTQIVQQTMDDG	753
Novel_Thiodiazotropha	QYSSRQLETMTGLKTHVVDQTQTEYNATKQEIDVKMNEIFQIADKLKELNTLLAKNSSTVT	763
Ruditapes_philippinarum	SYVAEQKVSVDLQSEVTSLKDDCCSKHQEMETIAARLIELMGESKQICLQMENCTAKKT	754
Chamelea_gallina	THTTQQTEGIDKLQNHVTSCLKDEYNSRKQEIIDEIKARLAQMEEENHNVSIEKITVSNT	756
Mercenaria_mercenaria	THTAEQTESIEKLQNVTSCLKDEYNSRKQEIDKIKARLAQMEEENHNVSIEKIAMSET	756

Mya_arenaria	THTRVQEERITGLRESLASLTDNYSNRHQIIEQKMSLIQLSQENHGIVGQMSSEVESRT	754
Dreissena_polymorpha	EHTRQQGQLIGDLHVTIETLTQAYNSRHKCIQDRLQELVQMTTDNQAVVSQISQQSVSVT	797
Human_KIF11	-----QKFCADS----DGFSQELRNFNQEGTK----LVEESVKHSDKLN---GNL	808
Sinanodonta_woodiana	QDRLT-----SASAVQCQFSSKVEEMKSLSQTHSQSIIDSSRCTCTQTKKEMDKA	803
Novel_Thiodiazotropha	QHSNHTLSEIEKSSSSAEDMSSNLNSEINHAKERSNSCLQESQEKLESNCQCVHKDIDKI	823
Ruditapes_philippinarum	-----ADIKTMAESMNNTTSVKREQILTNNERNLKEFYSVRDNHHNILQGEMNNV	804
Chamelea_gallina	-----ENLKSQARTTCNKTGEKRKHIEAENQSNLDEFSKIQEKHCKYSQEIQINI	806
Mercenaria_mercenaria	-----EDLKTKVKTTCTKTGEKRKHIDIENQSSLEEFFSIQEKQCRYSQEQIQINI	806
Mya_arenaria	-----KELCSETGSHLSSVQEWKSQTEKENTSKVEAIIIEQKLQHTRSASTRISEV	804
Dreissena_polymorpha	-----SHMRNEIKGHLAGMQQLRDTVEAENQTEVADIVQLKTNHCTATSGHIQQV	847
Human_KIF11	EKISQETEQRCESLNTRTVY-----FSEQWVSSL----NEREQELHNLLEVVSQCCEA	857
Sinanodonta_woodiana	DNLVNGCHSLCDSSLSQAKEYHQAATKAWETHCTDLQEKCKKHSDIKTDLDIHQNCTDG	863
Novel_Thiodiazotropha	KSVNTTVDIQTEIQTYNTVGKQRIKFLDSYQDTLEDRCTEHETTINTDLNVFYKNTDA	883
Ruditapes_philippinarum	EEVSQRMQEQCGHLKQSFMSCTSDSTEIVGRYSEKTSASTTQHITDINTGLNTHYKHTDD	864
Chamelea_gallina	KESSQRVQEECAQMKDSITTCNETTEITTYSESTATSCTQQVTDINTELNTHYKHTDN	866
Mercenaria_mercenaria	KKSSQNVQKECTEMKKSISNCNNEATDITTYHECTTTACTQQVTDINTELNTHYKHTDS	866
Mya_arenaria	QAIATKTHEDCTTVKTDINEHVTAGQTSQDQYTERLSDSCTQHVTDLNTALNTHNTHTQA	864
Dreissena_polymorpha	QTTSKNAHALCSAVHSDVQERVSKSRAALEKYSQETATTCTQQITTFNTQLNTHHSKTDA	907
Human_KIF11	SSSDITEKSDGRKAAHEKQHNIFLDQMTIDEDKL---IAQNL----ELNETIKIGLTKLN	910
Sinanodonta_woodiana	LTEECTKRCDSMVAIFKENKSLFEEDVAGIEQSLEERLQSTESMAAVFSSALVERSADVE	923
Novel_Thiodiazotropha	LQSENQACDKLISSLDSTASFDEAVATENQLLISRISGVRNFSEDQSARLTDRLSDVQ	943
Ruditapes_philippinarum	LQSVQSQWDRDLVDSVDKNSACVANEVETERSSTEERLSATSSYIEKQLNRLQEHKQKME	924
Chamelea_gallina	LQSTCQSQCDRLVDITDKQNASVDKEIEVEKTSAAEQGLATSYIENHRSKLQEHKLEMK	926
Mercenaria_mercenaria	LQSTCQTQCDRLVDITDKQNASVDKEIEVEKNSAAEQALTTSYVESQRSKLEEHKQEMR	926
Mya_arenaria	LETSVQRQCCKLVDVTDQRSADLDGVDKTERGCLNWLQGATAQSAAQFMEGLNQNRNEDMR	924
Dreissena_polymorpha	LETSMQLQCDRLVDITDKQGAADFEEVDREKTTVSSLRDRTLGFSSDLLARLEDNKEQMR	967
Human_KIF11	CFLEQDLKLDIPTGTTQPKRSYLYPSTLVRTEPREHLLDQLKRKQPELLMMLNCSENNKE	970
Sinanodonta_woodiana	KFIEDELKKDIPTGLTPVSKEVLYPQKLTQTDHHSILLNQYRELMKEAIE--EADDEMER	981
Novel_Thiodiazotropha	RFVGEELRKDIPTGQTPVRKDFQYPRDLKKTGRHSLLEKLFREDNMVNLSLDEDSIVEES	1003

Ruditapes_philippinarum	RYIKEEMSECVPTGATPQRQQFVIKRNLKRTSEHFTLLDRFRCDNMIDIE-EELDSDEL	983
Chamelea_gallina	RYIKEEMSECVPTGATPQRREFQIKRNLKRTSDHGTLLDRFRENMVDIE-EEIDSDIEQ	985
Mercenaria_mercenaria	RYIKEEMSECVPTGATPQRREFHIQRNLKRTSDHGTLLDRFRVDNMINIE-EEIDSDLEQ	985
Mya_arenaria	RYIRQEMSQDKPTGLTPVRHDFPHKRNVRISDQQKLLDRFRSDNMVEIEEEPVDSDTEL	984
Dreissena_polymorpha	RYIRQEISEDKPTGLTPVRSDYPHKRNIRISDQQKLLERFRAENPVSDIQEEPDSDEL	1027
Human_KIF11	ETIP-DVDVEEAVLGQYTEEP-----LSQEPSVDAG----VDC-----	1003
Sinanodonta_woodiana	RDLF-----SIPRRQNVQSA-----DEERDDGSDAASEKSYNSTTSG---	1018
Novel_Thiodiazotropha	DEIK---ETEM-----QNDDVADGKG-----	1021
Ruditapes_philippinarum	DSQN---QVKIVIDKTEDGNVIPDDKSDGINDDKSDRINDDKSDAG---SVKSSTSG---	1034
Chamelea_gallina	DSPC---QVKVVVDGAEDG-----QIADDKSDPG---SVKSSTSG---	1020
Mercenaria_mercenaria	DSQC---QVQVVVEAAEDDG-----QIADNKSDPG---SVKSSTSG---	1020
Mya_arenaria	ENVN--AQCLVVVHGPDNEQ-----G-----IDTSGHSDNI---SVKSSTSGISV	1024
Dreissena_polymorpha	ENMDTAKKCSIVIQGPNEQ-----LDVSDKSDTV---SVKSSTSG---	1065
Human_KIF11	-----SSIGGVFFQHK--K--SH---GKDKENRGINTLERSKVEE-----TTEHLVT	1044
Sinanodonta_woodiana	-----ISATSGFSALSDFSKA--SETKENVQKPGRPIGTLRKGKTTSQKAANVNTTPRG	1070
Novel_Thiodiazotropha	-----	1021
Ruditapes_philippinarum	-----VS---GLSSVSAASKV--SEYKE-NDRQGRPLGCKKS-----KKNFNNVTPRG	1076
Chamelea_gallina	-----VS---GLSSVSTASKV--SECKE-NDRQGRPLGAKKA-----KKNFANVTPRG	1062
Mercenaria_mercenaria	-----VS---GLSSVSAASKM--SECKE-NDRQGRPLGAKKS-----KKNFANVTPRG	1062
Mya_arenaria	KSSTSGVSEMSG-SAFSTTSKRSNLDTKE-NERQGRPLGNKKM-----KKNFMNVTPR-	1075
Dreissena_polymorpha	-----YSEISGLSAVSGVSKRSNLECKE-NDRQGRPIGNKKQ-----KKNFINVTPR-	1111
Human_KIF11	KSRLPLRAQINL-----	1056
Sinanodonta_woodiana	KTKLPLKTTNTNAEAN---	1086
Novel_Thiodiazotropha	-----	1021
Ruditapes_philippinarum	RTRLPLRQNNSTDN-----	1090
Chamelea_gallina	RTRLPLRQNNQAAE----	1077
Mercenaria_mercenaria	RTRLPLRQNNIEAAE----	1077
Mya_arenaria	KTRLPLANKQSNAEQQKTE	1094
Dreissena_polymorpha	KTRLPLRQANEQKIEEES-	1129

[Q6]: Phylogenetic Tree

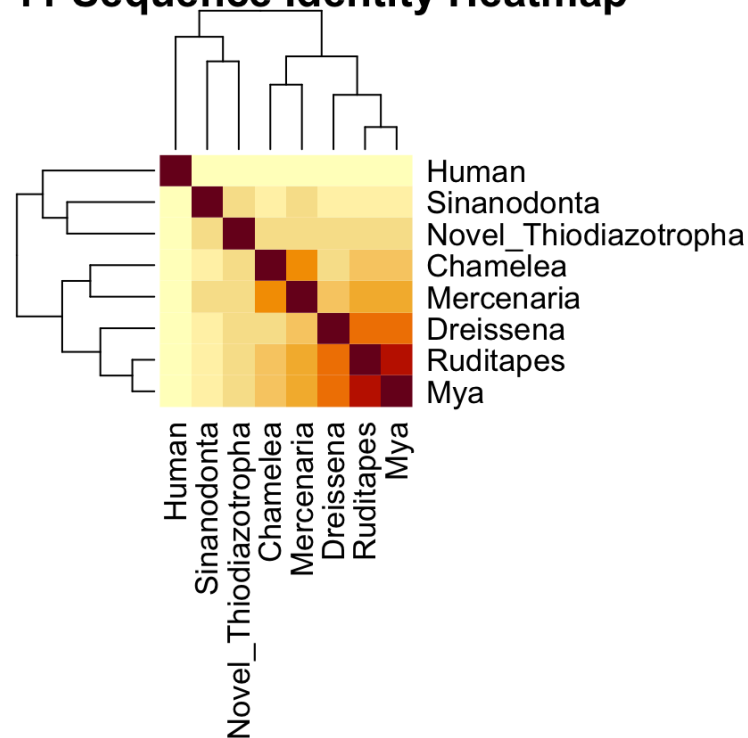
Figure A. Phylogenetic tree of KIF11-family proteins. A distance based phylogenetic tree generated from a multiple sequence alignment using Cluster Omega containing the novel protein from Candidatus Thiodiazotropha endolucinida, human KIF11, and six homologous KIF11-family proteins from diverse species.



[Q7]: Heatmap

All proteins included are homologs of KIF11 identified through BLAST analysis and multiple sequence alignment. The proteins included were Human KIF11, the novel *Candidatus Thiodiazotropha endolucinida* kinesin-like protein, and homologous KIF11-family proteins from *Sinanodonta woodiana*, *Mya arenaria*, *Mercenaria mercenaria*, *Chamelea gallina*, *Ruditapes philippinarum*, and *Dreissena polymorpha*.

KIF11 Sequence Identity Heatmap



[Q8]: Top Unique Hits

Top Unique Hit: PDB structure 6VPO is a cryo-electron microscopy structure of the microtubule-bound KLP61F motor domain in the AMPPNP state from *Drosophila melanogaster*. The structure was solved using electron microscopy (Cryo-EM) at a resolution of 4.4 Å. The hit matched the novel *Candidatus Thiodiazotropha endolucinida* kinesin-like protein with an E-value of 2e-148 and 61.81% sequence identity.

2nd Top Unique Hit: PDB structure 5MM4 is a cryo-electron microscopy structure of the *Mycosarcoma maydis* kinesin-5 motor domain bound to microtubules. The structure was solved using electron microscopy (Cryo-EM) at a resolution of 4.5 Å. The hit matched the novel *Candidatus Thiodiazotropha endolucinida* kinesin-like protein with an E-value of 6e-112 and 57.45% sequence identity.

3rd Top Unique Hit: PDB structure 5M5I is a pseudo-atomic model of the microtubule-bound *Schizosaccharomyces pombe* kinesin-5 motor domain in the AMPPNP state, based on a cryo-electron microscopy experiment. The structure demonstrates how the N-terminal conformation allows formation of a cover neck bundle. The structure was solved using electron microscopy (Cryo-EM) at a resolution of 9.3 Å. The hit matched the novel *Candidatus Thiodiazotropha endolucinida* kinesin-like protein with an E-value of 3e-111 and 51.09% sequence identity.

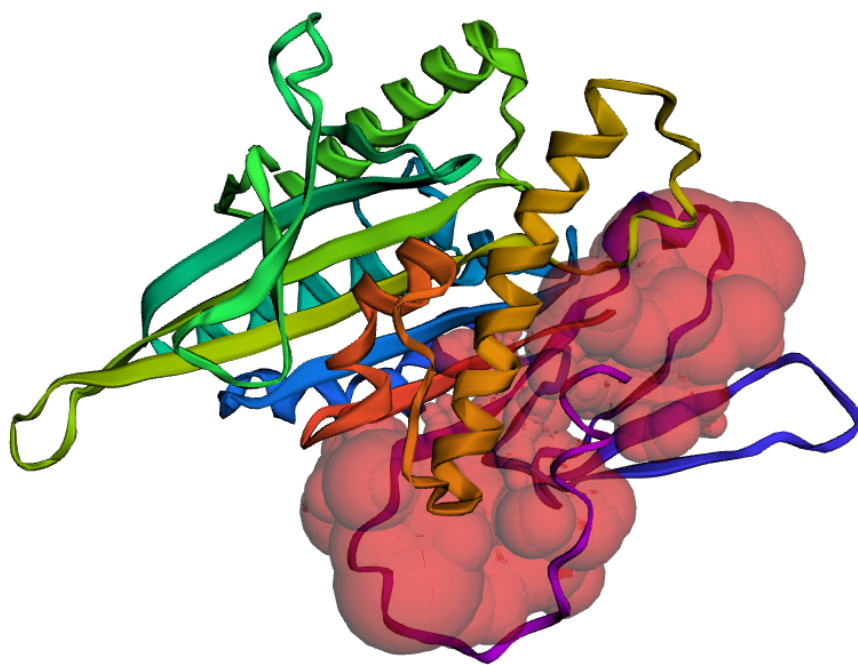
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☐	Chain K. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	506	506	32%	3e-169	66.58%	391	6TA3_K
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	504	504	32%	7e-169	66.40%	382	3WPN_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	503	503	32%	1e-168	66.67%	369	3HQD_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	502	502	32%	3e-168	66.67%	369	4ZCA_A
☐	Chain A. KINESIN-RELATED MOTOR PROTEIN Eg5 [Homo sapiens]	Homo sapiens	501	501	32%	4e-168	66.58%	368	1I6_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	501	501	32%	6e-168	66.40%	369	4ZHL_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	501	501	31%	6e-168	67.31%	367	1Q0B_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	501	501	32%	6e-168	66.40%	370	8YHH_A
☐	Chain A. KINESIN-LIKE PROTEIN KIF11 [Homo sapiens]	Homo sapiens	501	501	32%	7e-168	66.58%	370	4AP0_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	501	501	31%	9e-168	67.31%	367	6TLE_A
☐	Chain A. KINESIN-LIKE PROTEIN KIF11 [Homo sapiens]	Homo sapiens	500	500	32%	1e-167	66.58%	368	4A28_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	499	499	31%	2e-167	67.41%	359	1X88_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	499	499	31%	2e-167	67.04%	367	6TRL_A
☐	Chain A. KINESIN-LIKE PROTEIN KIF11 [Homo sapiens]	Homo sapiens	498	498	32%	1e-166	66.30%	368	4A1Z_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	498	498	32%	1e-166	66.30%	368	6Y1L_A
☐	Chain C. KINESIN-LIKE PROTEIN KIF11 [Homo sapiens]	Homo sapiens	498	498	32%	2e-166	65.94%	373	4AQV_C
☐	Chain A. KINESIN-LIKE PROTEIN KIF11 [Homo sapiens]	Homo sapiens	497	497	32%	3e-166	66.30%	368	4B7B_A
☐	Chain A. Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	495	495	31%	1e-165	67.99%	367	5ZOT_A
☐	Chain A. KINESIN-LIKE PROTEIN KIF11 [Homo sapiens]	Homo sapiens	484	484	30%	2e-161	67.72%	348	3ZCW_A
☒	Chain C. Kinesin-like protein Kip61F [Drosophila melanogaster]	Drosophila melan...	451	451	32%	2e-148	61.81%	377	6VPO_C
☐	Chain C. BIPOLAR KINESIN KRP-130 [Drosophila melanogaster]	Drosophila melan...	450	450	32%	3e-148	61.98%	373	2WBE_C
☐	Chain A. Kinesin-like protein Kip61F [Drosophila melanogaster]	Drosophila melan...	448	448	32%	2e-147	61.88%	369	7LXR_A
☒	Chain K. kinesin-5 [Mycosarcoma maydis]	Mycosarcoma ma...	355	355	27%	6e-112	57.45%	385	5MM4_K
☒	Chain C. Kinesin-like protein cut7 [Schizosaccharomyces pombe 972h-]	Schizosaccharom...	353	353	31%	3e-111	51.09%	369	5M5I_C
☐	Chain K. kinesin-5 [Mycosarcoma mauriel]	Mycosarcoma ma...	355	355	27%	6e-111	57.45%	457	5MM7_K



[Q9]: AlphaFold2 3D Structure



Figure B. AlphaFold2-predicted structure of a putative kinesin-like ATPase motor domain protein from *Candidatus Thiodiazotropha endolucinida* (GFOJ010595215.1_5). The protein is displayed as a cartoon colored by pLDDT confidence score, with conserved residues corresponding to the ATP-binding motif AYGQTGTGKT highlighted as spacefill spheres. The presence of this highly conserved motif and the overall predicted structure support its annotation as a kinesin-like ATPase motor domain protein.



(i) **Figure C.** CASTpFold analysis of the putative kinesin-like ATPase motor domain protein from *Candidatus Thiodiazotropha endolucinida*. The largest predicted binding pocket was Pocket 1, with a surface area of $2575.527 \text{ \AA}^2$ and a volume of $3210.052 \text{ \AA}^3$. This pocket may represent a potential ATP or ligand-binding region because it overlaps with the conserved ATP-binding motif identified in the protein sequence.

Search results for *Candidatus Thiodiazotropha endolucinida* GFOJ010595215.1_5

COMPOUNDS 0 TARGETS 0 ASSAYS 2 DOCUMENTS 4 CELL LINES 0 TISSUES 0

2 results

Tools: RELATED ACTIVITIES, CSV, TSV

View as cards

Showing 1-2 out of 2 records

Assay ChEMBL ID: CHEMBL3080392

Description: Selectivity ratio of LC50 for *Diaphorina citri* to LC50 for *Diaphorina citri* infected with *Candidatus Liberibacter asiaticus*

Assay Type: Functional

Assay ChEMBL ID: CHEMBL3080394

Description: Insecticidal activity against *Diaphorina citri* infected with *Candidatus Liberibacter asiaticus* assessed as mortality after 48 hr

Assay Type: Functional

(ii) A target search of ChEMBL was performed using the novel protein from *Candidatus Thiodiazotropha endolucinida* (GFOJ010595215.1_5). No matching ChEMBL target was identified (0 targets returned), and therefore no target-associated assays or ligand efficiency data were available. The assay results returned by the keyword search were unrelated to the novel protein and corresponded to other organisms containing the term "Candidatus." As a result, target-associated assay and ligand efficiency data were not available as of June 8, 2026.

(iii) The novel protein appears moderately druggable due to the presence of a large predicted binding pocket identified by CASTpFold (surface area = 2575.527 Å²; volume = 3210.052 Å³). Conservation analysis revealed a highly conserved ATP-binding motif (AYGQTGTGKT), suggesting a functionally important binding site that may be susceptible to inhibition. No target-associated assays or ligand efficiency data were available in ChEMBL for this specific protein, although related kinesin-like ATPases are known drug targets in other organisms. If this protein plays an essential cellular role, inhibitors targeting its conserved ATP-binding region could potentially disrupt its activity.

